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# Self-assembled, bivalent aptamers on graphene oxide as an efficient anticoagulant†

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Graphene oxide (GO) has unique structural properties, can effectively adsorb single-strand DNA through π–π stacking, hydrogen bonding and hydrophobic interactions, and is useful in many biotechnology applications. In this study, we developed a thrombin-binding-aptamers (15- and 29-mer) conjugated graphene oxide ($TBA_{15}$/$TBA_{29}$–GO) composite for the efficient inhibition of thrombin activity towards the formation of fibrin from fibrinogen. The $TBA_{15}$/$TBA_{29}$–GO composite was simply obtained by the self-assembly of $TBA_{15}$/$TBA_{29}$ hybrids on GO. The high density and appropriate orientation of $TBA_{15}$/$TBA_{29}$ on the GO surface enabled $TBA_{15}$/$TBA_{29}$–GO to acquire an ultrastrong binding affinity for thrombin (dissociation constant = $2.9 \times 10^{-12}$ M). Compared to bivalent $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ hybrids, the $TBA_{15}$/$TBA_{29}$–GO composite exhibited a superior anticoagulant potency (*ca.* 10-fold) against thrombin-mediated coagulation as a result of steric blocking effects and a higher binding affinity for thrombin. In addition, the prolonged thrombin clotting time, prothrombin time (PT), and activated partial thromboplastin time (aPTT) of $TBA_{15}$/$TBA_{29}$–GO were at least 2 times longer than those of commercially available drugs (heparin, argatroban, hirudin, and warfarin). The *in vitro* cytotoxicity and hemolysis analyses revealed the high biocompatibility of $TBA_{15}$/$TBA_{29}$–GO. The rat-tail bleeding assay of the hemostasis time and *ex vivo* PT and aPTT further revealed that $TBA_{15}$/$TBA_{29}$–GO is superior (>2-fold) to heparin, which is commonly used in the treatment and prevention of thrombotic diseases. Our multivalent, oligonucleotide-modified GO nanocomposites are easy to prepare, cost-effective, and highly biocompatible and they show great potential as effective anticoagulants for the treatment of thrombotic disorders.

## Introduction

Hemostasis is a multi-stage blood clotting phenomenon that naturally occurs in the event of damage to blood vessels and tissues to prevent blood loss.[1] Clotting involves a series of enzymatic reactions, mostly in the blood vessels and the cell surfaces, that result in the formation of a fibrin clot.[2] However, the formation of blood clots inside a blood vessel, a condition called thrombosis, disrupts the cardiovascular system and even the normal functioning of some organs.[3] Hence, the coagulation cascade has been extensively studied to understand the mechanism at different stages of hemostasis to treat or prevent thrombosis.[4] Thrombin (Factor IIa) is a serine protease belonging to the chymotrypsin family and has many important roles in the coagulation cascade.[5] The main function of thrombin is the conversion of soluble fibrinogen into insoluble fibrin strands.[6] In addition, thrombin performs various other functions, such as feedback amplification of the coagulation factors V, VIII, XI and XIII. It also activates protease-activated receptors on platelets, leading to platelet activation and aggregation.[7] Therefore, inhibiting the activity of thrombin is one of the most efficient thrombosis treatment methods, and it is of great interest in the development of anticoagulants.[8]

Thrombin has two electropositive exosites, *i.e.*, the anion-binding exosites (ABE)-I (the fibrinogen-binding site) and ABE-II (the heparin-binding site), in addition to its catalytic site.[9] Thrombin utilizes either or both ABEs to interact with its substrates and cofactors for the development of thrombosis.[10]

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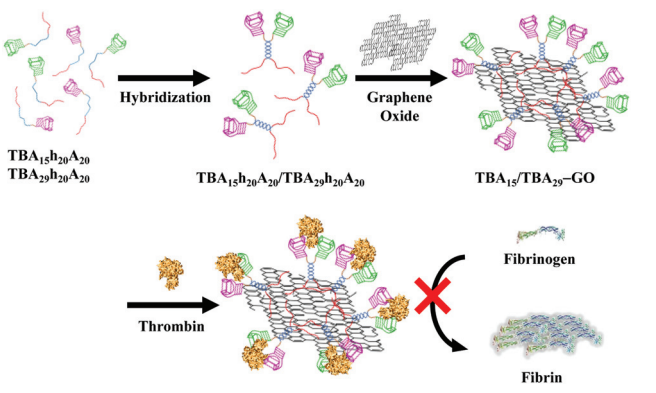
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Many drugs have been developed for directly or indirectly inhibiting thrombin activity through binding to the active site and/or ABE-I and ABE-II of thrombin.[11] Heparin, which is a widely used anticoagulant, indirectly inhibits the free thrombin activity by simultaneously binding to antithrombin and ABE-II.[12] However, heparin cannot inhibit the activity of fibrin-bound thrombin.[12] Moreover, heparin can non-specifically interact with various plasma proteins, causing an undesirable anticoagulant response, including side effects such as heparin-induced thrombocytopenia (HIT).[12]

Direct thrombin inhibitors (DTIs), such as lepirudin, desirudin, bivalirudin, argatroban, and dabigatran, directly bind to both soluble thrombin, fibrin-bound thrombin and fibrin degradation product-bound thrombin to exert their inhibition ability without the need for a cofactor.[13] Therefore, compared to heparins, DTIs do not cause immune-mediated thrombocytopenia and exhibit a more predictable anticoagulant effect since most of them do not bind to plasma proteins.[13] Some DTIs such as lepirudin and argatroban have been employed for prophylaxis and treatment of venous thromboembolism (VTE) and prevention of thromboembolic complications in patients with HIT or at risk for HIT.[14] Although some DTIs exhibit efficient anticoagulation effects at high doses for reducing the risk of stroke and systemic embolism in nonvalvular atrial fibrillation and for the treatment of deep venous thrombosis and pulmonary embolism, their short plasma half-lives (<1 h) and side effects, such as pain in the back or pelvis, sour stomach, nausea, nervousness, blurred vision and bleeding, limit their clinical uses.[15] Accordingly, developing a new type of anticoagulant that is effective with few side effects is highly desired.

Aptamers are single-stranded oligonucleotides with specific 3-dimensional structures that allow them to bind to target proteins or small molecules with a high affinity and specificity.[16–19] Two types of thrombin-binding aptamers (TBAs), a 15-base nucleotide ($TBA_{15}$) and a 29-base nucleotide ($TBA_{29}$), have been developed through systemic evolution of ligands by exponential enrichment (SELEX), and they form stable G-quadruplex structures to specifically bind to ABE-I and ABE-II of thrombin, respectively.[20,21] The dissociation constants ($K_d$) of $TBA_{15}$ and $TBA_{29}$ with their respective binding sites in thrombin are 100 and 0.5 nM, respectively, and their binding affinity is weak.[20,21] Only $TBA_{15}$ has potential anticoagulation and anti-thrombin-mediated platelet activation and aggregation effects because of its interaction with the fibrinogen-binding ABE-1 of thrombin.[20] However, due to its low binding affinity, $TBA_{15}$ is required in very high concentrations to achieve an appropriate anticoagulant response.[22] A number of bivalent and multivalent TBAs have been developed by anchoring TBAs on nanoparticles and polymers and constructing TBA-tagged DNA origami to improve the binding affinity of TBAs for thrombin and their anticoagulation potencies.[23–30] However, to anchor the TBAs on polymers and nanoparticles requires expensive and tedious functionalization processes.[24–27] Previously, we developed a self-assembly of triblock aptamers on gold nanoparticles for the effective inhibition of thrombin activity on fibrinogen.[31] However, the aptamers adsorbed on the Au NP surfaces are easily replaced by the highly concentrated aminothiols in human blood, which reduces their affinity for thrombin. Furthermore, the expense and metallic toxicity of Au NPs also limit their clinical applications.[32] Thus, the anticoagulant abilities of the reported TBA-modified polymers and metallic nanoparticles and TBA-tagged DNA origami have only been proven *in vitro*.[23–30] Therefore, easier, more cost-effective methods with highly biocompatible nanomaterials are needed to develop new TBA-conjugated anticoagulants.

In this study, we prepared simple TBA–graphene oxide ($TBA_{15}$/$TBA_{29}$–GO) nanoconjugates to enhance the binding of TBAs with thrombin to improve the anticoagulant activity (Scheme 1). GO is known to preferentially interact with single-stranded (ss) nucleic acids with poly(adenine) sequences.[33–36] Hence, the addition of a poly(adenine) sequence can improve the immobilization of TBAs on the GO surface. We used hybrid TBAs ($TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$) with poly(adenine) at the 3′ or 5′ terminus (see Table S1 for their sequences, ESI†) for anchoring on GO. The studied sequences are $TBA_{15}h_{20}A_{20}$: 5′-TCA-GTG-G-$TBA_{15}$-TG-CTT-GA-TTTTT-$h_{20}$-$A_{20}$-3′ and $TBA_{29}h_{20}A_{20}$: 5′-$A_{20}$-$h_{20}$-TTTTT-$TBA_{29}$-3′. We optimized these sequences based on our previous findings.[37] $TBA_{29}$ has been reported to adopt a G-quadruplex structure with a loop and stem in the terminal, while $TBA_{15}$ does not.[20,21] Therefore, two 7-mers were inserted before and after $TBA_{15}$ unit to provide a loop and stem (G-quadruplex) structure for increasing the stability of $TBA_{15}$ aptamer. In addition, $T_5$ linker has been inserted to provide same length and enough spatial separation between $TBA_{15}A_{20}h_{20}$ and $TBA_{29}A_{20}h_{20}$ for bivalent binding with thrombin, enabling high-efficiency anticoagulation. The poly(adenine) parts of the hybrid TBAs adsorb on GO through multivalent π–π stacking, hydrogen bonding, and van der Waals interactions to produce stable $TBA_{15}TBA_{29}$–GO composites.[33–35] $TBA_{15}$/$TBA_{29}$–GO exhibited an ultrahigh stability in plasma and a good binding affinity for thrombin. The *in vitro* cytotoxicity and hemolysis and *in vivo* toxicity evaluations revealed the high biocompatibility of $TBA_{15}$/$TBA_{29}$–GO.




**Scheme 1** schematic representation of the self-assembly of thrombin-binding aptamer hybrids on GO ($TBA_{15}$/$TBA_{29}$–GO) and the effective inhibition of the thrombin-mediated catalytic reaction of fibrinogen to fibrin.

Thromboelastography (TEG) of whole blood coagulation, *in vivo* tail bleeding in a rat model, and *ex vivo* prothrombin time (PT) and activated partial thromboplastin time (aPTT) assays were performed to demonstrate the anticoagulation efficiency of $TBA_{15}$/$TBA_{29}$–GO.

## Results and discussion

### Self-assembly of hybridized TBAs on GO

The GO used in this study was prepared by a modified Hummer's method, a procedure previously reported by Marcano *et al.*[38] GO was prepared from graphite (7–11 μm) powder and exfoliated by ultrasonication (please see the Experimental section for details). The atomic force microscopy (AFM) measurements indicated that the single-layer GO has a thickness and average size of *ca.* 1.2 and 250 nm, respectively (Fig. S1A, ESI†). Jasim *et al.*, demonstrated that thin layer (~5 nm) and small sized-GO (lateral size <500 nm) are more easily excreted through the glomerular filter into the urine in rodents.[40,41] Therefore, the as-synthesized GO could be well suited for biomedical applications. The elemental analysis (EA) revealed that the carbon, hydrogen, and oxygen compositions of the as-synthesized GO were 62.79%, 3.93% and 32.66%, respectively, which was consistent with the results from previous reports.[43–45] We designed $TBAh_{20}A_{20}$ with three regions, a sequence for the TBA functionality, a 20-base sequence for hybridization ($h_{20}$), and a 20-repeat polyadenine ($A_{20}$) for anchoring. The hybrid $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ self-assembled on the GO to form $TBA_{15}$/$TBA_{29}$–GO conjugates through the strong interaction of single strand $A_{20}$ DNA with GO.[33–36,46–48] GO has long-range order in the $sp^2$ carbon lattice with a hydrophobic character and surface with oxygen-containing polar functional groups.[49] The ssDNA chains preferentially anchored laterally on the surface of the GO through π–π stacking, hydrophobic interactions, and hydrogen bonding.[33–35] The electrostatic repulsive interaction between the hybridized region (double strand DNA region) of the $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ complexes and GO facilitates the stable G-quadruplex structures of the TBAs which protrude out of the GO for effective multivalent interactions with thrombin.

We estimated the adsorption density of the TBAs on GO in the $TBA_{15}$/$TBA_{29}$–GO conjugates using OliGreen, and it was found to be 2.1 ± 0.3 nmol $mg^{-1}$ ($n$ = 3), which suggested a higher density of TBA ligands on the surface of the GO compared to the reported results for the saturated adsorption of poly-adenine on GO (~2.3 nmol $mg^{-1}$).[50] The height of $TBA_{15}$/$TBA_{29}$–GO measured by AFM was *ca.* 13.4 nm (Fig. S1B†), which indicated a monolayer of $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ was on both sides of the GO based on the size and length of TBA (~2.0 nm) and $h_{20}$ (~6.8 nm; 0.34 nm $bp^{-1}$).[51] The average thickness and particle size of $TBA_{15}$/$TBA_{29}$–GO were ~1.3 and 250 nm, respectively, which are below the accepted particle size limit for intravenous injection for biomedical applications.[52,53] We conducted the dynamic light scattering (DLS) measurements of GO and $TBA_{15}$/$TBA_{29}$–GO in various conditions. The results suggest that the anchoring of the $TBA_{15}$/$TBA_{29}$ hybrid on the surface of GO increased its dispersibility significantly. Compared to bare GO, the self-assembled $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ on the GO exhibited higher stability even in the highly saline physiological buffer and two-fold diluted human plasma samples (Fig. S2, ESI†). In addition, the stability of $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$–GO hybrid, $TBA_{15}h_{20}A_{20}$–GO and $TBA_{29}h_{20}A_{20}$–GO in two-fold diluted human plasma was determined by quantifying the DNA released to the solution using the OliGreen method. The amount of $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ hybrids released from GO after incubation in the 2-fold diluted human plasma was determined to be less than 1%. In contrast, more than 30% of $TBA_{15}h_{20}A_{20}$ or $TBA_{29}h_{20}A_{20}$ was released from $TBA_{15}h_{20}A_{20}$–GO or $TBA_{29}h_{20}A_{20}$–GO under the same conditions, which showed that $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ possesses a stronger stability on GO in biofluids. Previous studies revealed that many plasma proteins have moderate affinity [dissociation constant ($K_d$) ~ $10^{-6}$–$10^{-7}$ M] for GO, which can lead to non-specific binding.[54] However, the steric effects of the high-density G-quadruplex structures of the $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ hybrids on GO and the cooperative binding of the two $A_{20}$ sequences from the TBA hybrids with GO decrease the possibility of plasma proteins to access the basal planes of GO and replace the TBA hybrids.

### Comparison of the inhibitory activity of the free TBAs and $TBA_{15}$/$TBA_{29}$–GO

We studied the anti-thrombin activity of the free TBAs and $TBA_{15}$/$TBA_{29}$–GO by clotting tests using total concentrations of $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ were held constant (50 nM) in these experiments. Coagulation occurs through thrombin-mediated cleavage of two peptides, fibrinopeptides a and b, of fibrinogen, resulting in the formation of fibrin.[55] The binding knobs "A" and "B" in the central nodule of the fibrinogen monomer turn the complementary holes "a" and "b" in the α- and β-nodules into two-stranded protofibrils, and then, the protofibrils laterally aggregate to form thicker fibers.[55] We studied the real-time kinetics of coagulation by recording the scattering light intensity as a function of time (Fig. 1A). The activity of thrombin induces the formation of a larger fibrin gel, resulting in scattering of light. We compared the inhibition efficiencies of the free TBAs and $TBA_{15}$/$TBA_{29}$–GO for thrombin activity based on their initial reaction rates (within 200 s), and higher rates represent a weaker inhibition ability (Fig. 1B). The results showed that only $TBA_{15}h_{20}A_{20}$ exhibited a much stronger inhibitory potency (0.21 kcps $s^{-1}$) in comparison with that of $TBA_{29}h_{20}A_{20}$ (0.55 kcps $s^{-1}$) because $TBA_{15}$ binds to the critical fibrinogen-binding site of thrombin.[56] The inhibition induced by the free $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ hybrids was 2.5- and 6.3-fold stronger than that of $TBA_{15}h_{20}A_{20}$ and $TBA_{29}h_{20}A_{20}$, respectively, because bivalent TBAs ($TBA_{15}$ and $TBA_{29}$) possess a stronger binding affinity for thrombin.[57–61]

Among the tested aptamers and conjugates, $TBA_{15}$/$TBA_{29}$–GO (0.027 kcps $s^{-1}$) exhibited the highest anticoagulation efficiency (Fig. 1B) due to the multivalent interaction between

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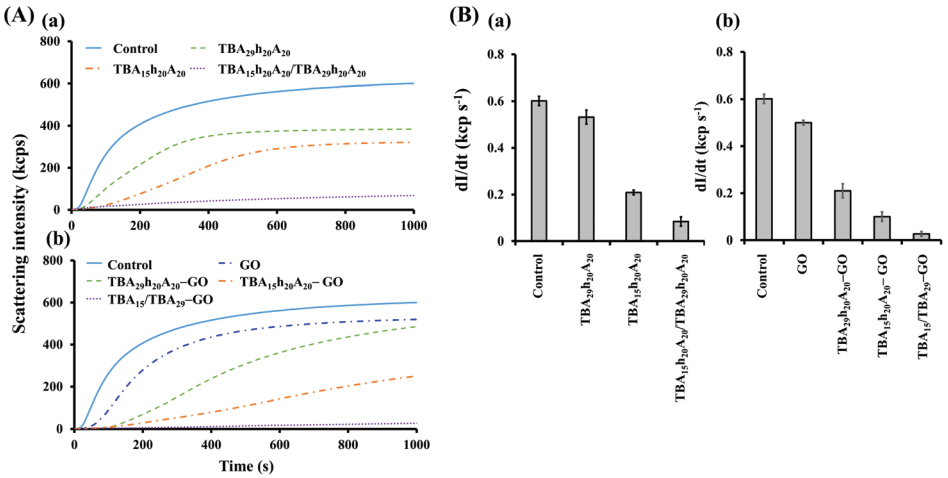



**Fig. 1** (A) Intensity of light scattering as a function of time to monitor the coagulation rate of mixtures of thrombin (7.5 nM), fibrinogen (1.14 μM), BSA (100 μM), and (a) free TBAs (50 nM) or (b) TBA-modified GO ([TBA] = 50 nM). (B) Initial reaction rates ($dI/dt$) obtained from the light scattering measurements for the coagulation process in the presence of (a) free TBAs (50 nM) or (b) TBA-modified GO ([$TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$] = 50 nM). Error bars in (B) represent the standard deviation of three repeated measurements.

$TBA_{15}/TBA_{29}$–GO and thrombin and the efficient blocking of the thrombin exosites and/or active sites for fibrinogen. Additionally, the appropriate orientation of the $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ hybrids on the GO surface and the desired distance between $TBA_{15}h_{20}A_{20}$ and $TBA_{29}h_{20}A_{20}$ resulted in efficient multivalent binding. Compared with free $TBA_{15}$ ($K_d$ ~ 100 nM) and $TBA_{29}$ ($K_d$ ~ 0.5 nM),[20,21,56] $TBA_{15}/TBA_{29}$–GO exhibited a higher binding affinity towards thrombin ($K_d$ = 2.9 × $10^{-12}$ M; Fig. S3, ESI†). Moreover, the binding affinity of $TBA_{15}/TBA_{29}$–GO for thrombin was also better than that previously reported for bivalent $TBA_{15}$–$TBA_{29}$ ($K_d$ = $10^{-9}$–$10^{-11}$ M).[57–61] The enhanced binding affinity of $TBA_{15}/TBA_{29}$–GO could be due to the ultrahigh density of $TBA_{15}/TBA_{29}$ on the GO surface and their appropriate orientation, which resulted in apparent, high, local concentrations of binding TBA ligands. In addition, other factors such as steric effects and electrostatic interactions between thrombin and GO may also contribute to the high inhibitory potency of $TBA_{15}/TBA_{29}$–GO. In addition, GO also slightly inhibited thrombin activity in the presence of a background protein (BSA, 100 μM) (Fig. 1b). Thrombin could outcompete BSA for binding to GO due to its stronger binding affinity for GO ($K_d$ = 2.0 × $10^{-10}$ M; Fig. S4, ESI†) in the presence of 100 μM BSA. The affinity of thrombin for GO is approximately 3 × $10^4$-fold higher than that of serum albumin ($K_d$ ~ 6 × $10^{-6}$ M).[62] Compared to covalently anchored aptamer molecules on GO assume flattened structures,[63–67] which cause a weak affinity for target molecules. However, the $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ self-assembled on the surfaces of GO exhibited a high flexibility and appropriate spatial orientation, which allowed the nanoconjugates to strongly interact with thrombin.

### Anticoagulation activities in plasma and blood

From the above results, it is clear that $TBA_{15}/TBA_{29}$–GO inhibited thrombin activity in biological media. To understand applications in real samples, we further studied the activity in human plasma samples by thrombin clotting time (TCT) tests, prothrombin time (PT), and activated partial thromboplastin time (aPTT). TCT tests are commonly performed to diagnose a bleeding and/or clotting disorder and as screening test for coagulation factors I, IIa, and XIII in the common coagulation pathways.[68] PT assays can evaluate the extrinsic and common pathways and determine factors II, V, VII, and X, and screening tests of aPTT are for factors II, V, VIII, IX, X, XI, and XII of the intrinsic and common pathways.[69] The results of the TCT assay for the anticoagulant activity of $TBA_{15}/TBA_{29}$–GO is presented in Fig. S5 (ESI†). The concentration-dependent TCT displayed in Fig. 2 clearly demonstrated that $TBA_{15}/TBA_{29}$–GO had the highest inhibitory function compared to that of the free $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ hybrids and the commercial anticoagulant drugs heparin, argatroban, hirudin, and war-

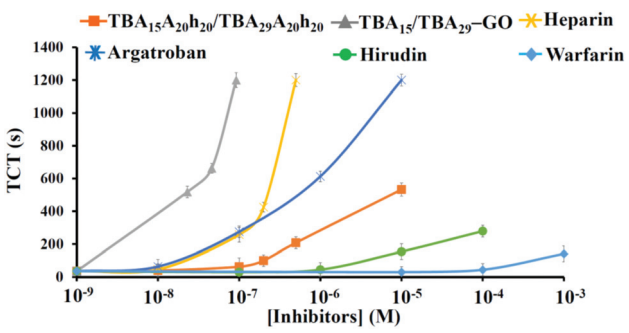



**Fig. 2** Dose-dependence of the TCTs of the $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ hybrids, $TBA_{15}/TBA_{29}$–GO conjugates and the commercial drugs heparin, argatroban, hirudin, and warfarin in human plasma samples. The TCT for each sample was noted as the time at which the scattering signal intensity was halfway between the lowest and highest points. The longest monitoring time of the clotting assays was set at 1200 s. Error bars represent the standard deviation of three repeated measurements.

farin. The TCT delay caused by $TBA_{15}/TBA_{29}$–GO ([$TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$] = 100 nM) was ~10 and >4 times longer than that caused by the $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ hybrids and commercial anticoagulant drugs (100 nM), respectively. The anticoagulant potency of $TBA_{15}/TBA_{29}$–GO ([$TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$] = 50 nM) was four times higher than that of the $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ hybrids (50 nM) in the buffer system (Fig. 1). The higher antithrombin activity of $TBA_{15}/TBA_{29}$–GO in plasma could be due to the presence of antithrombin III (AT III), which forms a ternary complex with thrombin and GO.

Fig. 3 shows a comparison of the prolonged PT and aPPT assays with $TBA_{15}/TBA_{29}$–GO, the $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ hybrids and four commercial drugs. The $TBA_{15}/TBA_{29}$–GO exhibited at least a 2 times higher anticoagulation potency than the $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ hybrids and commercial anticoagulant drugs on the basis of the prolonged PT and aPTT results. Our results suggested that $TBA_{15}/TBA_{29}$–GO significantly delayed the coagulation triggered by the common, intrinsic and extrinsic pathways relative to the other anticoagulants. In addition, $TBA_{15}/TBA_{29}$–GO is highly stable in plasma; therefore, it did not exhibit any significant change in the TCT delay even after 48 h of incubation in human plasma (Fig. S6, ESI†). The high local salt concentrations on the surfaces of $TBA_{15}/TBA_{29}$–GO and steric blocking endow the TBAs with the ability to resist nuclease digestion in plasma.[70] The major problems associated with using free oligonucleotide aptamers are their short half-lives in plasma, typically a few minutes, which make the drug delivery difficult; and the need for higher concentrations. Therefore, structural modification of the aptamers or conjugation with a suitable host molecule or nanostructured material is essential to prolong their half-lives, for the successful clinical application.

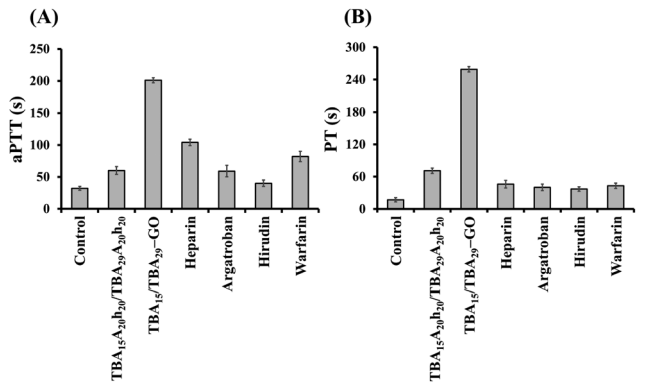
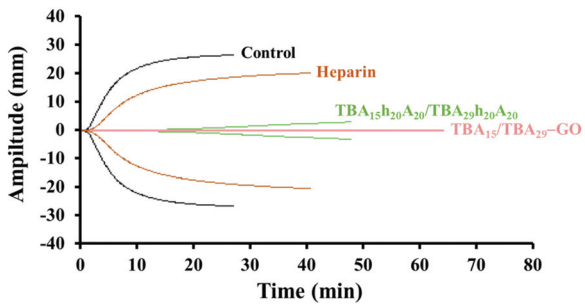



**Fig. 3** (A) PT and (B) aPTT of the $TBA_{15}/TBA_{29}$–GO conjugates, $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ hybrids, and the commercial drugs heparin, argatroban, hirudin, and warfarin in human plasma. The concentrations of TBA or commercial drugs in the PT and aPPT assays were 0.5 μM. To calculate the PT and aPTT, the end time was chosen as the point at which the scattering signal was halfway between the lowest and highest points. Error bars represent the standard deviation of three repeated measurements. Other conditions were the same as those described in Fig. 2.

Platelets have many functions in the formation of blood clots, including aggregating to provide a surface for coagulation factor assembly, physically incorporating into the fibrin clot and releasing internal agonists to amplify the coagulation responses.[71] The TCT, aPTT and PT assays still have some limitations in evaluating the efficiency of $TBA_{15}/TBA_{29}$–GO since they are performed in plasma in the absence of platelets (thrombocytes). Therefore, we investigated the thrombin inhibition property of $TBA_{15}/TBA_{29}$–GO with thrombin-activated thromboelastography (TEG) to determine the kinetics of clot formation inhibition in whole blood.[72] The mechanical motion in the TEG assay was used to quantify the blood clot formation, including the lag time (*R* time; time of latency from the start of the test to the initial fibrin formation), time taken to achieve a certain level of clot strength (*K* time; amplitude of 20 mm), angle ($\alpha$ angle; the rate of clot formation), and maximum amplitude (MA; the mechanical strength of the clot formed).[72] As shown in Fig. 4, $TBA_{15}/TBA_{29}$–GO exhibited a superior anticoagulation potency compared to that of the $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ hybrids and heparin (most widely used anticoagulant) based on the *R* time, *K* time, $\alpha$ angle, and MA values. The results further demonstrated the advantages of $TBA_{15}/TBA_{29}$–GO over heparin with respect to the TEG parameters. Although many multivalent TBA-based nanoconjugates have been prepared to improve the anticoagulant activity of TBAs, their anticoagulation efficiency in whole blood has not been reported.[23–30] Our results clearly reveal that $TBA_{15}/TBA_{29}$–GO nanoconjugates exhibit a strong anticoagulant activity in whole blood.

## Biocompatibility

We evaluated the cytotoxicity and *in vitro* hemolytic effect of $TBA_{15}/TBA_{29}$–GO towards mammalian cells and red blood cells (RBCs), respectively. The cytotoxicity of $TBA_{15}/TBA_{29}$–GO for different mammalian cells was assessed using the XTT cell proliferation assay (Fig. S7, ESI†). The A549 (human lung




| | R (min) | K (min) | Angle (deg) | MA (mm) |
|---|---|---|---|---|
| Control | 1.2 ± 0.6 | 2.0 ± 0.4 | 62.4 ± 0.2 | 55.0 ± 0.4 |
| $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ | 20.9 ± 0.5 | N/A | 2.7 ± 0.5 | 5.8 ± 0.6 |
| $TBA_{15}/TBA_{29}$–GO | >65 | N/A | N/A | N/A |
| Heparin | 2.0 ± 0.3 | 4.4 ± 0.6 | 41.0 ± 0.6 | 41.0 ± 0.4 |

**Fig. 4** Effects of the $TBA_{15}/TBA_{29}$–GO conjugates ([$TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$] = 0.5 μM), $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ hybrids (0.5 μM), and heparin (0.5 μM) on thrombin-activated TEG. N/A: not available.


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adenocarcinoma epithelial cell line), MCF-7 (human breast adenocarcinoma cell line), HeLa (human cervix adenocarcinoma cell line), and HEK-293 T (human embryonic kidney 293 cell line) cells did not show a significant cytotoxic effect (cell viability >95%) after incubation with $TBA_{15}/TBA_{29}$–GO for 24 h at a concentration of 0.4 mg $mL^{-1}$ ([$TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$] = 1.0 μM), which is much higher than the effective concentration of $TBA_{15}/TBA_{29}$–GO for inhibiting thrombin in plasma (Fig. 2 and 3) and whole blood (Fig. 4). Furthermore, an *in vitro* hemolysis assay was performed to verify the biocompatibility of $TBA_{15}/TBA_{29}$–GO since anticoagulants are widely used in blood-related therapy to prevent and/or treat thrombotic disorders. Defibrinated human blood was used for the hemolysis assay to evaluate the viability of fragile RBCs in the presence of different concentrations of $TBA_{15}/TBA_{29}$–GO in a PBS solution. $TBA_{15}/TBA_{29}$–GO did not cause significant hemolysis of RBCs at concentrations up to 0.4 mg $mL^{-1}$ ([$TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$] = 1.0 μM) (Fig. S8, ESI†). The *in vitro* biocompatibility evaluations revealed that $TBA_{15}/TBA_{29}$–GO has substantial potential as a safe anticoagulation drug because of its high thrombin inhibition. In our *in vivo* and *ex vivo* experiments, the dosage of GO entering the rat through the intravenous administration of $TBA_{15}/TBA_{29}$–GO is ~0.8 mg per kg body weight, which is significantly below the dose of GO reported for pathological changes in rats.[73] However, for clinical applications, further studies using animal models to determine the pharmacokinetic parameters, biodistribution, metabolism, and chronic toxicity should be conducted.

### *In vivo* and *ex vivo* anticoagulant evaluations

A rat-tail transection model was employed to assess the antihemostatic effect of $TBA_{15}/TBA_{29}$–GO *in vivo*. The rats (~200–250 g) were dosed by intravenous injection with heparin (0–4.0 μM; 100 μL) or $TBA_{15}/TBA_{29}$–GO ([$TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$] = 0–4.0 μM; 100 μL). After 5 min, the rat tails were fully transected 2 mm from the tip. The control rats with a dose of physiological buffer (100 μL) had an average hemostasis time of 6.9 ± 1.2 min ($n$ = 5), as shown in Fig. 5. The hemostasis time of the rat tail dramatically increased as the concentration of $TBA_{15}/TBA_{29}$–GO increased. $TBA_{15}/TBA_{29}$–GO was significantly superior to heparin in prolonging the hemostasis time at concentrations higher than 1.0 μM. The body weights of the tested rats 10 days after the dose of $TBA_{15}/TBA_{29}$–GO ([$TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$] = 4 μM, 100 μL) were almost the same as that of the untreated ones ($p > 0.05$, $n$ = 5; data not shown). In addition, all rats treated with $TBA_{15}/TBA_{29}$–GO survived for 2 months. We further conducted *ex vivo* PT and aPTT assays to evaluate the blood anticoagulant activity of $TBA_{15}/TBA_{29}$–GO. The rat plasma samples were collected after 30 min of intravenous administration of $TBA_{15}/TBA_{29}$–GO or heparin. As shown in Fig. S9 (ESI†), the PT and aPTT values for the $TBA_{15}/TBA_{29}$–GO-treated group increased to 60.5 ± 3.4 s and 65.2 ± 6.5 s (mean ± SD, $n$ = 5), while there was no obvious increase in the PT and aPTT values in the heparin-treated group. The high *in vivo* and *ex vivo* anticoagulation potencies of $TBA_{15}/TBA_{29}$–GO indicate that it has great potential for the treatment of thrombotic diseases.

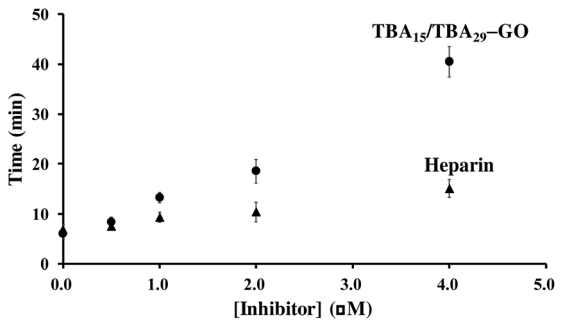



Fig. 5 The effect of $TBA_{15}/TBA_{29}$–GO conjugates and heparin on rat-tail bleeding times using different doses of the inhibitors (0–4.0 μM, 100 μL) administered *via* intravenous injection. Error bars represent the standard deviation of five rat measurements.

## Conclusions

We have developed a simple route to prepare a nanodrug, $TBA_{15}/TBA_{29}$–GO, which exhibits highly effective anticoagulation by specifically inhibiting thrombin activity towards fibrinogen. Compared with free TBA ($K_d \sim 10^{-7}$–$10^{-9}$ M) and TBA-modified polymeric and metallic nanoparticles ($K_d \sim 10^{-9}$–$10^{-11}$ M),[20,21,23–27] the high density of $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ ligands and their appropriate orientation on the GO surface play a crucial role in providing ultra-strong interactions ($K_d = 2.9 \times 10^{-12}$ M) with thrombin in ensuring superior anticoagulation properties. Moreover, most reported TBA-based nanoconjugates require complex and tedious procedure for anchoring aptamers on polymers or nanoparticles and are rarely employed for the anticoagulation *in vivo*.[23–27,37] The easily prepared, highly stable and biocompatible $TBA_{15}/TBA_{29}$–GO composite exhibited a much superior anticoagulant function in human plasma as well as *in vivo* rat tail-bleeding model than free $TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}$ hybrids and heparin, showing its potential for treating coagulation-related cardiovascular diseases. Our study reveals that different aptamers can be further immobilized on GO for synergistic anticoagulation by simultaneously inhibiting multiple coagulation factors and platelets.

## Experimental section

### Materials

The graphite powder (99%, 7–11 μm) for the synthesis of GO was obtained from Alfa-Aesar (Heysham, Lancashire, UK). Phosphoric acid and acetic acid were purchased from Mallinckrodt Baker (Phillipsburg, NJ, USA). Potassium permanganate, sulfuric acid and hydrogen peroxide were obtained from SHOWA (Tokyo, Japan). Sodium chloride, potassium

chloride, sodium phosphate, magnesium chloride, calcium chloride, human α-thrombin (≥1000 NIH units per mg of protein), fibrinogen, and bovine serum albumin (BSA) were obtained from Sigma-Aldrich (Milwaukee, WI, USA). Commercial anticoagulants, argatroban, warfarin and heparin (sodium salt; $M_w$ approximately 5000 Da) were purchased from Sigma-Aldrich, and hirudin was purchased from AnaSpec (Fremont, CA, USA). The oligonucleotides listed in Table S1† were purchased from Integrated DNA Technologies (Coralville, IA, USA). The PT and aPTT assay reagents were purchased from Helena Laboratories (Beaumont, Texas, USA). All other reagents were purchased from Sigma-Aldrich. Sprague-Dawley rats were purchased from BioLASCO (Ilan, Taiwan).

### Synthesis and characterization of graphene oxide

Graphene oxide (GO) was prepared using a modified version of Hummer's method.[38] Briefly, concentrated $H_2SO_4$ and $H_3PO_4$ were used in a 9 : 1 volume ratio (90 : 10 mL) and added to a mixture of graphite powder (0.75 g) and $KMnO_4$ (5 g). The mixture was heated to 50 °C and maintained at the same temperature for 12 h with constant stirring. The resulting mixture was allowed to cool to room temperature and placed in an ice bath. Then, 100 mL of deionized (DI) water followed by 3 mL of 30% $H_2O_2$ were added slowly to the mixture with continuous stirring. The as-formed graphite oxide was separated by centrifugation at a relative centrifugal force (RCF) of 35 000*g* for 30 min, and the supernatant was decanted. The obtained residue was then repeatedly washed with 200 mL of sodium phosphate buffer (5 mM, pH 12) until the pH reached ~7 and finally with DI water. To exfoliate the graphite oxide, it was suspended in 200 mL of DI water and sonicated for 2 h (power 200 W). The as-obtained GO dispersion was then centrifuged at an RCF of 15 000*g* for 0.5 h. The supernatant containing GO was collected, and the residue containing un-oxidized graphite and multilayered GO was discarded. The concentration of GO in the supernatant was determined by the freeze-drying method. The average size of GO was determined (~250 nm) using an atomic force microscope (Shimadzu SPM-9600 AFM, Shimadzu Co, Kyoto, Japan) with the purified sample deposited onto a mica sheet.

### Preparation of the self-assembled $TBA_{15}$/$TBA_{29}$–GO

The $TBA_{15}h_{20}A_{20}$ (2.5 μM) and $TBA_{29}h_{20}A_{20}$ (2.5 μM) were mixed with GO (0.08 μg $mL^{-1}$) in a 5 mM sodium phosphate buffer (pH 7.2) containing 100 mM sodium chloride and incubated at room temperature for 2 h. Then, the $TBA_{15}$/$TBA_{29}$–GO samples were salt-aged (with 1 M NaCl) for 2 h and centrifuged at a relative centrifugal force (RCF) of 55 000*g* for 2 h to remove excess $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$. The amount of $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ in the supernatant after centrifugation was measured using a single-strain DNA labeling dye, OliGreen, to determine the number of $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$ molecules adsorbed on GO. The fluorescence intensity of OliGreen exhibits over 1000-fold enhancement at 524 nm upon binding to oligonucleotides when excited at 480 nm. The precipitate was washed with physiological buffer (25 mM Tris-HCl (pH 7.4), 150 mM NaCl, 5 mM KCl, 1 mM $MgCl_2$, and 1 mM $CaCl_2$) with two centrifuge/wash cycles and stored at 4 °C.

### Measurement of the real-time coagulation kinetics

The inhibition efficiency of $TBA_{15}$/$TBA_{29}$–GO ([$TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$] = 50 nM) for thrombin activity was assessed by a light scattering experiment. Either free TBAs (50 nM) or $TBA_{15}$/$TBA_{29}$–GO ([$TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$] = 50 nM) were added to aliquots (500 μL) of physiological buffer containing 100 μM BSA and fibrinogen (1.14 μM) and allowed to stand for 15 min. Thrombin (7.5 nM) was then added, and light scattering measurements were performed using an FP-6500 spectrophotometer (JASCO, Tokyo, Japan) at excitation and emission wavelengths of 650 nm. The coagulation rate can be directly obtained by measuring the scattered light intensity. Hence, the coagulation rate was calculated from the linear portion of the early slope (0–200 s) of the profiles. This initial rate represents the relative thrombin-inhibiting strength of the inhibitor.

### TCT assay

The anticoagulation potency of the free TBAs and $TBA_{15}$/$TBA_{29}$–GO for the common coagulation pathway were assessed by TCT assays. In addition, the anticoagulation efficiency of $TBA_{15}$/$TBA_{29}$–GO was also compared with that of four commercial anticoagulants (heparin, warfarin, hirudin and argatroban). The human plasma for the TCT assays was collected from healthy individuals according to the rules and research ethics of the institution. For the TCT experiments, the inhibitor was first incubated with two-fold diluted human plasma at 37 °C in a physiological buffer (475 μL) for 3 min. Then, thrombin (15 nM, 25 μL) was added to initiate the clotting cascade. The intensity of the scattered light was recorded at 650 nm. The thrombin clotting time was noted as the point where the intensity of the scattered light was halfway between its lowest and highest values. Three repeated measurements were obtained for each set of experiments using a single batch of plasma.

### PT and aPTT assays

The anticoagulation properties of $TBA_{15}$/$TBA_{29}$–GO were further evaluated by determining the prothrombin time (PT) and activated partial thromboplastin time (aPTT) delay in human plasma samples. For the PT assay, $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$, $TBA_{15}$/$TBA_{29}$–GO or commercial inhibitors (1 μM, 300 μL) were separately incubated with 200 μL of human plasma at 37 °C for 120 s, followed by the addition of 150 μL of PT reagent for the initiation of the extrinsic clotting cascade. The coagulation was monitored by measuring the intensity of the scattered light until the signals reached saturation. For the aPTT determinations, 160 μL of human plasma was incubated with 440 μL of each inhibitor at 37 °C for 10 min, and then, 100 μL of aPTT reagent was added and incubated for 2 min. To initiate the intrinsic clotting cascade, 100 μL of pre-warmed $CaCl_2$ solution was added. The aPTT assay was monitored using a method similar to that of the PT

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assay. To calculate the PT and aPTT, the end time was chosen as the point at which the scattered light signal intensity was halfway between the lowest and maximal values. Three repeated measurements were obtained for each set of experiments using a single batch of plasma.

### Determination of the binding constant of thrombin with $TBA_{15}$/$TBA_{29}$–GO

To understand the enhanced affinity of $TBA_{15}$/$TBA_{29}$–GO for thrombin, we determined the dissociation constant ($K_d$) for the $TBA_{15}$/$TBA_{29}$–GO and thrombin interaction. Briefly, aliquots of each $TBA_{15}$/$TBA_{29}$–GO solution ([$TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$] = 100 pM) with different concentrations of thrombin (0–80 pM) were mixed in a physiological buffer in the presence of 100 μM BSA and incubated for 30 min to form thrombin–$TBA_{15}$/$TBA_{29}$–GO complexes. The thrombin–$TBA_{15}$/$TBA_{29}$–GO complexes and unbound thrombin were separated by centrifugation at an RCF of 50 000*g* for 60 min. The unbound thrombin in the supernatant was estimated using fibrinogen–modified Au NPs, as reported earlier.[39] The saturation binding data were processed using the Scatchard equation below to examine the binding properties.

$$N_{\text{Thrombin}}/[\text{Free-Thrombin}] = N_{\max}/K_d - N_{\text{Thrombin}}/K_d$$

where $N_{\text{Thrombin}}$ is the number of thrombin molecules bound to $TBA_{15}$/$TBA_{29}$–GO at equilibrium, [Free-Thrombin] is the free thrombin concentration at equilibrium, $N_{\max}$ is the apparent maximal number of binding sites, and $K_d$ is the dissociation constant. A graph was plotted with $N_{\text{Thrombin}}$/[Free-Thrombin] against $N_{\text{Thrombin}}$ to obtain a linear relationship. The slope and intercept of the plot represent $1/K_d$ and $N_{\max}/K_d$, respectively.

### Thromboelastography

The anticoagulation efficiency of $TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$, $TBA_{15}$/$TBA_{29}$–GO and heparin in whole blood clots was evaluated through the viscoelastic development of thrombin-mediated fibrin polymerization and platelet activation using a thromboelastograph analyzer (Haemoscope corporation, Niles, IL, USA). Blood samples were collected from a healthy volunteer (24 years old) into tubes containing sodium citrate (3.2%), and the blood collection procedures were performed in compliance with the relevant laws and institutional guidelines. The plain disposable plastic TEG cups (Haemonetics) were equilibrated at 37 °C before the experiments. First, 0.5 μM of thrombin inhibitor (52 μL) in a physiological buffer was mixed with 288 μL of whole blood and incubated in the TEG cup at 37 °C for 10 min. Then, 20 μL of a 15 nM thrombin solution was added to the TEG cups to initiate the whole blood coagulation. The clot formation at 37 °C was measured with a thromboelastograph analyzer until a stable clot formed or one hour had passed. The time lag time, $\alpha$ angle, and maximum amplitude were automatically calculated by the TEG® Analytical Software version (TAS) 4.2.3 (Haemonetics).

### *In vitro* cytotoxicity

To establish the biocompatibility of $TBA_{15}$/$TBA_{29}$–GO, its *in vitro* cytotoxicity effect on human lung adenocarcinoma epithelial cells (A549), human breast adenocarcinoma cells (MCF-7), human cervical cancer cells (HeLa), and human embryonic kidney epithelial cells (HEK-293T) was studied. All cell lines were obtained from the American Type Culture Collection (ATCC, Manassas, VA, USA). The A549, MCF-7, HeLa and HEK-293T cell lines were cultured in alpha-MEM, L-15, DMEM and DMEM, respectively, supplemented with fetal bovine serum (10%), antibiotic–antimycotic (1%), L-glutamine (2.0 mM), and nonessential amino acids (1%) in 5% $CO_2$ at 37 °C. Prior to the cytotoxicity test, approximately $2 \times 10^5$ cells per well of all the cell lines were separately incubated with their respective culture medium (5% $CO_2$) in 24-well plates at 37 °C for 8 h. Then, the culture medium in each well was replaced with 400 μL of a medium containing different concentrations of $TBA_{15}$/$TBA_{29}$–GO ([$TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$] = 0–1.0 μM) and incubated for 24 h. The total number of viable cells was determined by a 2,3-bis-(2-methoxy-4-nitro-5-sulfophenyl)-2*H*-tetrazolium-5-carboxanilide (XTT) cell proliferation assay. After 24 h of incubation, the medium was removed from each well, and the cells were washed with a physiological buffer. A 1 mL XTT solution (0.3 mg $mL^{-1}$) was added and incubated at 37 °C under 5% $CO_2$ for 4 h. After this time, the XTT solution was removed, and deionized water (1 mL) was used to dissolve the resulting blue formazan crystals in the living cells. The optical density was determined at 492 nm with a Synergy H4 Multi-Mode Microplate Reader (BioTek Instruments, Winooski, VT, USA).

### Hemolysis assays

The red blood cells (RBCs) for the hemolysis assays were collected in accordance with the laws and research ethics of the institution. Blood samples were collected from a healthy volunteer (24 years old) into tubes containing ethylenediaminetetraacetic acid (EDTA) and centrifuged within 30 min at an RCF of 3000*g* for 10 min at 4 °C to remove the serum. The RBCs were diluted with sterile isotonic physiological buffer to obtain an RBC stock suspension (*ca.* 4 vol% of blood cells). For the analysis, 100 μL of the RBC stock suspension was incubated with aliquots of $TBA_{15}$/$TBA_{29}$–GO solutions ([$TBA_{15}h_{20}A_{20}$/$TBA_{29}h_{20}A_{20}$] = 0.1–1.0 μM) at 37 °C for 1 h in 1.5 mL vials. Then, they were centrifuged at an RCF of 1000*g* for 10 min. Hemolysis was measured based on the absorption of hemoglobin at 576 nm ($OD_{576}$) in the supernatant (200 μL). For the 0% hemolysis reference ($OD_{576\ \text{blank}}$), a sterile isotonic physiological buffer was used. The positive control, 100% hemolysis, was prepared by adding ultrapure water to an RBC suspension ($OD_{576\ \text{ultrapure water}}$). The hemolysis activity was calculated as follows:

$$\text{Hemolysis}\,(\%) = [(OD_{576\,\text{TBA15/TBA29–GO}} - OD_{576\,\text{blank}})/(OD_{576\,\text{ultrapure water}} - OD_{576\,\text{blank}})] \times 100$$

**Determination of rat-tail bleeding time and *ex vivo* PT and aPTT**

To study the applicability of the $TBA_{15}/TBA_{29}$–GO composite as an anticoagulant, *in vivo* studies were conducted on male Sprague-Dawley rats (200–250 g) using a rat-tail incision bleeding model with a slight modification to a previously reported procedure.[42] Permission for the rat animal studies was obtained from the Institutional Animal Care and Use Committee of the National Laboratory Animal Center (IACUC Approval number: 105021). The male rats (Sprague-Dawley rats, weighing 200–250 g) were anesthetized using Zoletil 50 at a dose of 100 μL per 100 g body weight *via* subcutaneous injection, and $TBA_{15}/TBA_{29}$–GO ($[TBA_{15}h_{20}A_{20}/TBA_{29}h_{20}A_{20}]$ = 0–4.0 μM, 100 μL) or heparin (0–4.0 μM, 100 μL) was administered by an intravenous tail injection. After 5 min, the rat-tail tip was cut at 2 mm, and the tails were immersed in physiological buffer. The time until the cessation of bleeding was recorded, and a statistical analysis was performed using Student's *t*-test. *P* values <0.05 were considered significant. A Kaplan–Meier method was used to determine the probability of rat survival.

For *ex vivo* PT and aPTT measurements, $TBA_{15}/TBA_{29}$–GO (10 μM, 200 μL), heparin (10 μM, 200 μL), or a physiological buffer control (200 μL) was intravenously administered to male Sprague-Dawley rats (200–250 g) for 30 min. The rats were then anesthetized, and their whole blood (2.7 mL) was collected by cardiac puncture and transferred into plastic tubes containing citrate (0.3 mL, 3.2%). The PT and aPTT values were determined under the conditions described above.

## Conflicts of interest

There are no conflicts to declare.

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